

OSCAR NDUGBU

Staff Backend and Platform Engineer · AI Infrastructure · Reliability

scardubu@gmail.com · scardubu.dev
github.com/Scardubu ·
linkedin.com/in/oscardubu
+234 803 388 5065 · Lagos, Nigeria · Open to Remote

PROFESSIONAL SUMMARY

Staff backend and platform engineer with independent product and consulting work across tax workflows, AI-assisted observability, and encrypted data systems — anchored by over a decade building and maintaining data infrastructure within Nigeria's federal public sector.

Core expertise spans **Node.js and TypeScript backend systems, distributed architecture, and cloud infrastructure** engineered around explicit failure modes, observable behavior, and clean operational handoffs. Applies machine learning where it improves a system decision, and translates technical trade-offs into clear business terms. Author of three public engineering tools.

CORE COMPETENCIES

Languages	TypeScript · Node.js · Python · SQL · Bash
Backend & APIs	Fastify · Express · FastAPI · GraphQL · REST · gRPC · BullMQ · WebSocket
Data & Cloud	PostgreSQL · Redis · Prisma ORM · Docker · Kubernetes · AWS · GCP · GitHub Actions
Observability / SRE	Prometheus · Grafana · OpenTelemetry · ELK Stack · Sentry · Distributed Tracing
AI / ML	XGBoost · LightGBM · LLM APIs (GPT-4, Claude) · FastAPI Inference · ZK-Proofs
Systems Design	Multi-Tenant RLS · RBAC · Event-Driven Architecture · Circuit Breaker · AES-256 · Zero-Downtime Deploys

PROFESSIONAL EXPERIENCE

Infrastructure Consultant — Backend & Reliability

Jan 2024 – Present

Independent Consulting · Remote

Retained by fintech and SaaS engineering teams to audit systems, resolve reliability issues, and build internal tooling — working directly alongside CTOs and senior engineers.

- **Built an AI-assisted incident diagnosis tool** that turns live server logs into ranked root-cause hypotheses and preserves the evidence path for engineer review.
- **Conducted backend performance audits** across fintech systems — identifying slow database queries, connection-pool exhaustion, and API latency spikes, then delivering written remediation plans tied to observable bottlenecks.
- **Designed proactive monitoring pipelines** that surface anomalous behavior with enough context for an engineer to investigate before a local fault becomes a wider incident.

Lead Backend Engineer — Encrypted Data Platform

Jan – Jun 2024 (contract via consulting practice)

Hashablanca · Remote · Ethereum, Polygon, BSC, StarkNet

Privacy-tech startup building infrastructure for the secure, verifiable transfer of sensitive data across multiple blockchain networks.

- **Designed and built the core transfer backend** for AES-256-encrypted payloads across four blockchain networks, with retry semantics, checkpointed progress, and a transaction audit record designed for recovery after interruption.
- **Integrated Zero-Knowledge proof pipelines** to validate data without exposing it, supporting privacy-preserving reporting with tests across the integration boundaries. This architecture is not presented as regulator certification.

Senior Data & Systems Engineer — Planning, Research & Statistics

2014 – Present

Universal Basic Education Commission (UBEC) · Abuja HQ · Lagos-based · Federal Government

UBEC is Nigeria's federal agency for basic education funding. The Planning, Research & Statistics department produces the national data and analysis used to allocate education budgets across all 36 states.

- **Designed and built automated ETL pipelines** that collect, validate, and consolidate education statistics from all 36 state offices into a single audit-ready database — replacing weeks of manual spreadsheet work each reporting cycle.
- **Developed interactive dashboards and policy reports** in Python and Power BI, giving senior planners a real-time view of national education metrics and directly supporting budget allocation decisions at the federal level.
- **Automated repeated validation and consolidation steps**, making the preparation path more traceable while preserving human review for inconsistent state submissions.

PRODUCTS BUILT

TaxBridge — SME Tax Compliance & Filing Platform

Fastify 5 · PostgreSQL 15 RLS · Redis 7 · BullMQ · GraphQL · Docker · GitHub Actions

- Designed a replay-safe tax filing workflow for Nigerian small businesses, with Nigeria Revenue Service integration in progress. PostgreSQL Row-Level Security enforces tenant scope at the database layer instead of relying only on application filters.
- Separates interactive calculation from external submission. Stable job identities, idempotent workers, and a hash-chained audit trail make retries and reconciliation inspectable after a partial failure.
- Uses encrypted storage, release automation, and executable tests across calculation and submission state transitions. Private integration evidence is available for review and is not presented as agency approval.

SabiScore — ML Prediction & Observability Platform

FastAPI · XGBoost / LightGBM / CatBoost · Redis Pub/Sub · Prometheus · Grafana

- An ensemble inference platform with versioned caching, delivery telemetry, and a deterministic fallback path when model output is unavailable.
- Separated inference work from repeated reads through a model-versioned cache; the event path uses bounded retries and dead-letter recovery instead of silently dropping failed work.
- Prometheus and Grafana expose model delivery, latency, and service-health signals so an engineer can trace a degraded response to its source.

OPEN SOURCE CONTRIBUTIONS

pg-tenant	<i>Node.js · PostgreSQL</i>	Provides database-level tenant scoping with PostgreSQL RLS, reducing reliance on application filters and making isolation tests explicit.
audit-chain	<i>Fintech · Compliance</i>	Generates a hash-linked audit sequence so a later edit breaks the verification chain at the affected record.
node-debug-llm	<i>AI · DevOps</i>	Turns live logs and traces into ranked root-cause hypotheses while keeping the underlying evidence available for engineer review.

Public repositories and authored engineering notes: github.com/Scardubu · scardubu.dev

LEADERSHIP & COMMUNITY

- **Architecture advisor** to early-stage fintech founders — reviewing system design decisions, ML deployment strategies, and security posture at critical pre-launch and early-growth stages.
- **Speaker at Lagos JS and Python community meetups** on distributed systems design, deploying ML models in production, and building resilient backend infrastructure.
- **Technical author at scardubu.dev** — publishing practical engineering guides on backend reliability, platform decisions, and AI infrastructure.

CERTIFICATIONS & EDUCATION

AWS Certified Developer · Dec 2023	GCP Associate Cloud Engineer · Aug 2023
OpenJS Node.js Services (JSNSD) · May 2024	PostgreSQL 14 Associate · Mar 2024

B.Tech, Environmental Technology · Federal University of Technology, Owerri · 2006–2011

Non-CS academic background. Certification details are resume-listed and available for verification on request; public repositories provide the inspectable engineering record.